

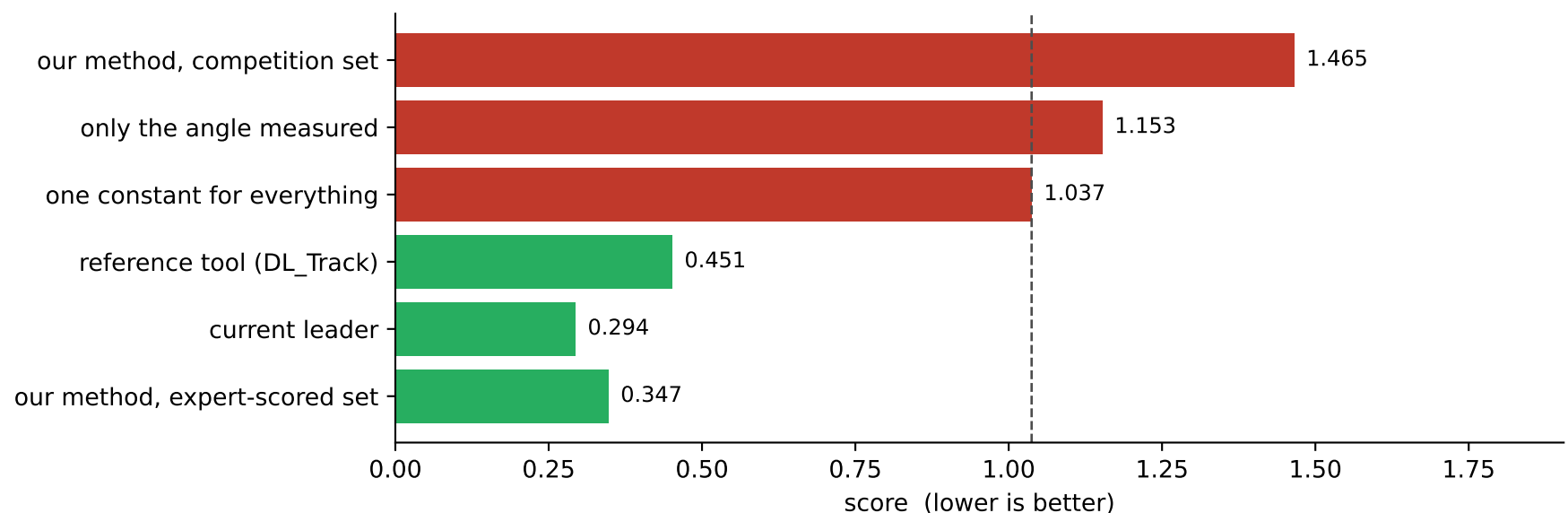
Why our muscle measurements fail on the competition images

UMUD Challenge — what we know, what we ruled out, what is still open

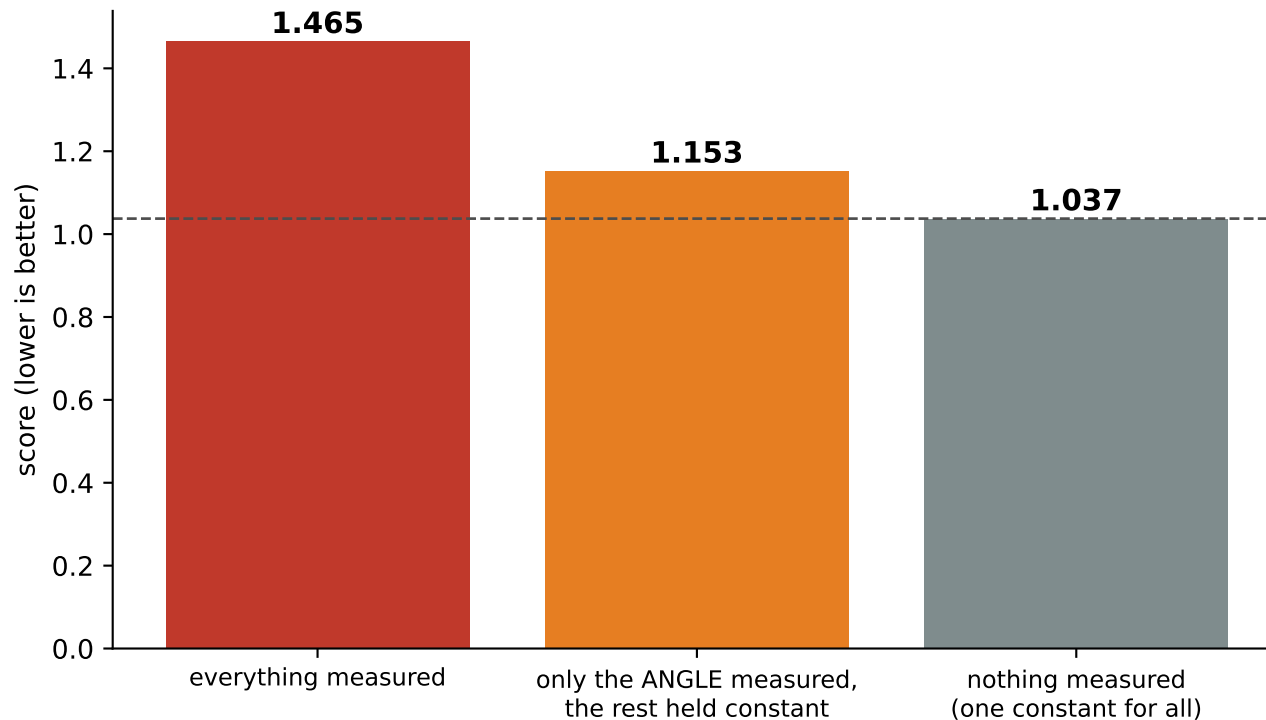
We measure three things on each ultrasound picture: the angle the muscle fibres make (pennation angle), how long they are (fascicle length), and how thick the muscle is. Lower score = closer to the experts.

On 35 pictures scored by seven human experts, our method gets 0.347 — as good as the published reference tool, and near the average human.

On the actual competition pictures it gets 1.465. Submitting the SAME number for every picture scores 1.037. So our measurements are currently worse than not measuring at all.



1. The measurement itself is the problem — proved by a controlled test

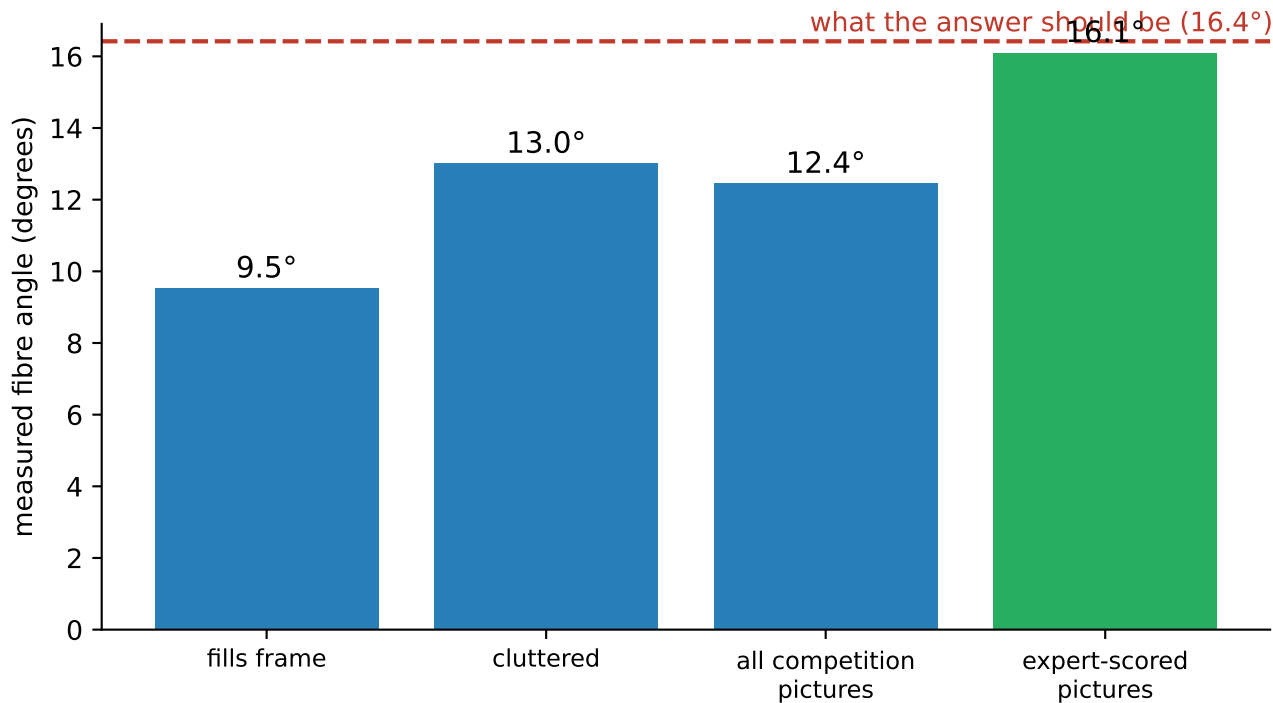


We submitted a version where the **ONLY** thing we measured was the fibre angle; fascicle length and thickness were replaced by a fixed number.

It scored worse than using fixed numbers for all three (1.153 vs 1.037).

That is the key result: our measured angle carries less information than a guess. On the 35 expert-scored pictures the same code is accurate to about 1.3 degrees. So the code is not simply imprecise — something about the competition pictures breaks it.

2. The angle comes out roughly 40% too small — consistently

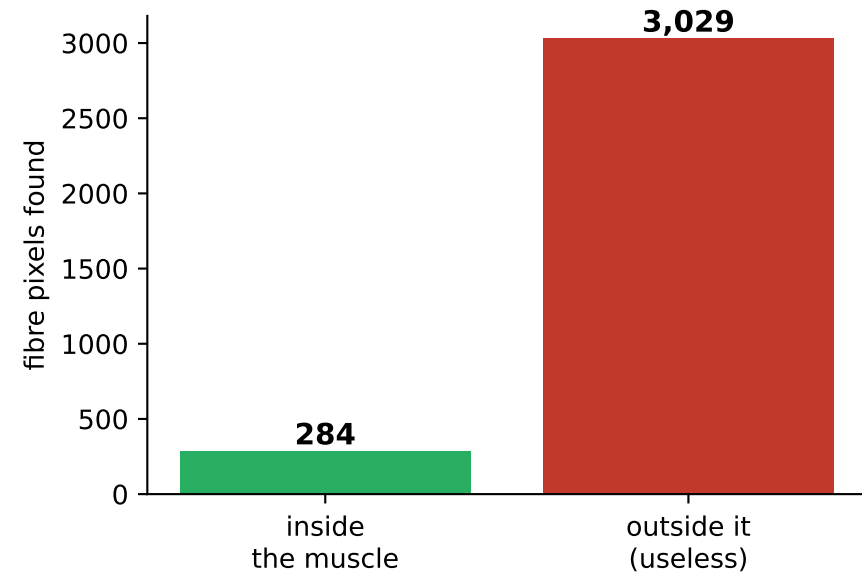
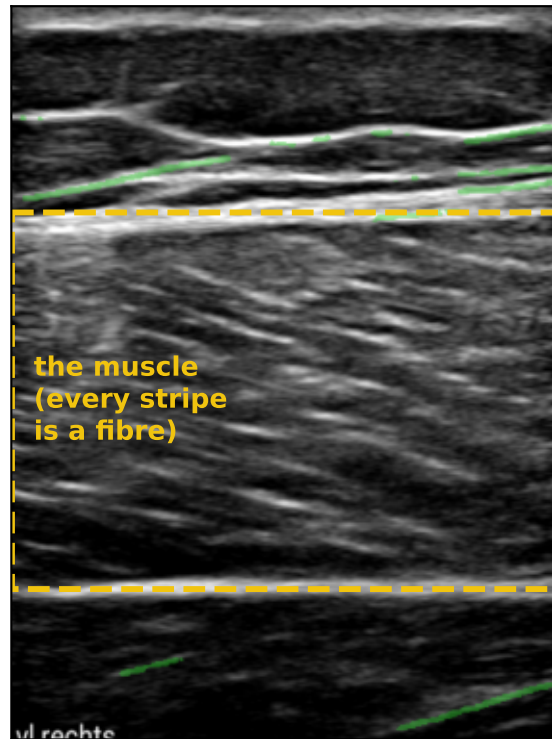


We know the correct middle value for the competition pictures: 16.4 degrees. We worked it out from the scoreboard itself, by submitting a few fixed numbers and watching how the score changed. The same method predicted the score of a constant submission to five decimal places, so it is reliable.

Our measurements land near 10-13 degrees. On the expert-scored pictures the same code is correct. Something is rotating or squashing the competition pictures relative to what we assume — or the fibres we are measuring are not the real ones.

3. One thing we can see going wrong: the fibres are found outside the muscle

IMG_00252.png
green = what the computer thinks are fibres

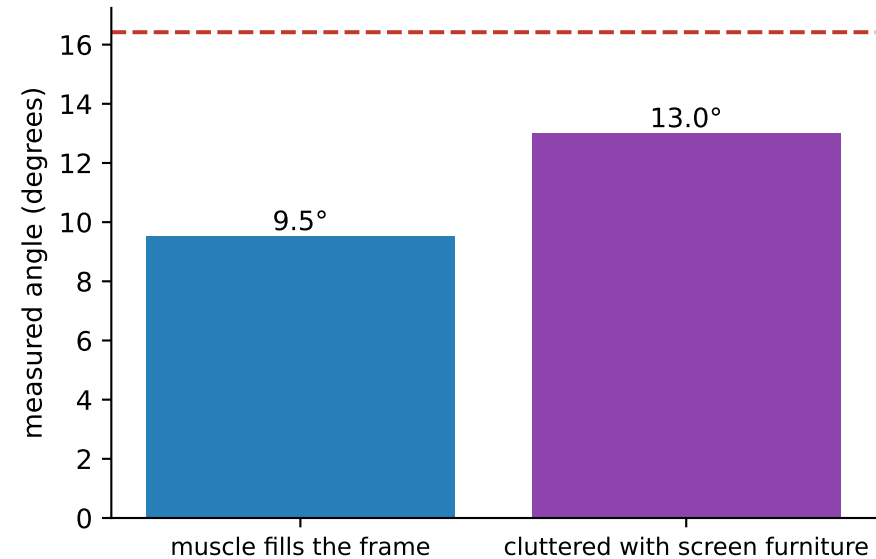
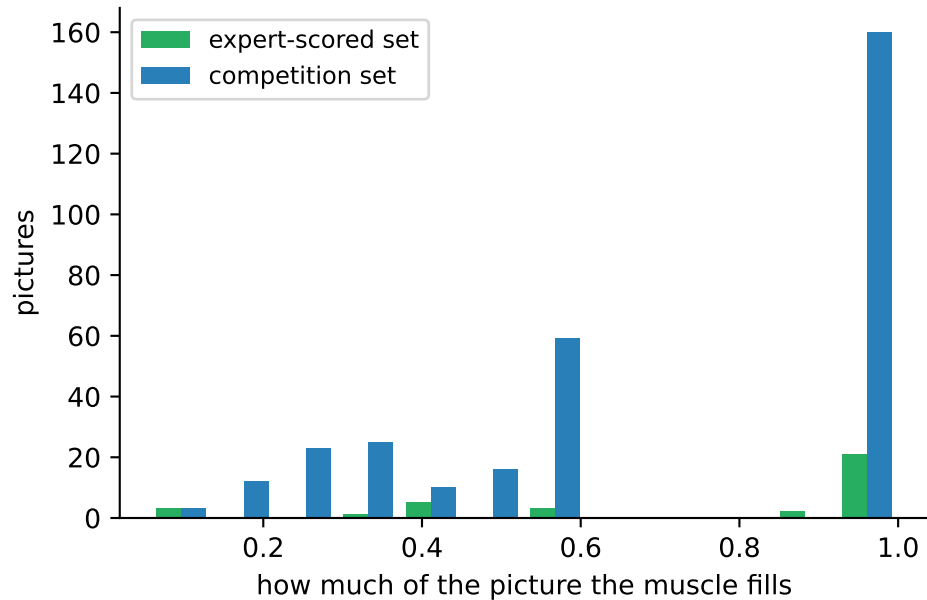


Inside the yellow box the muscle is covered in obvious diagonal stripes — those ARE the fibres we are asked to measure. The computer finds almost none of them, and puts 91% of its marks in the fat above and below the muscle instead.

Fibre angle and fibre length are computed from those marks, so on this picture they cannot be right.

Caveat: this is one picture, examined closely. We have not yet shown it explains the whole failure.

4. What we ruled out — and why that matters



Our first theory was that the competition pictures show the whole ultrasound machine's screen, so the muscle is small and gets squashed when the computer shrinks the picture to a fixed size.

Measured, that is wrong. The muscle fills 95% of the typical competition picture — MORE than in the expert-scored set (92%). And the pictures where the muscle fills the frame give a WORSE angle (9.5°) than the cluttered ones (13.0°), which is the opposite of what the theory predicts.

An earlier draft of this document asserted that theory. The measurement above is why it no longer does.

5. Where this leaves us

ESTABLISHED

- Our measurements score worse than fixed numbers on the competition set.
- The fibre angle specifically is worse than a guess, though the same code is accurate on the expert-scored pictures.
- The angle comes out about 40% too small, consistently.
- On at least one picture the fibre detector marks the wrong tissue entirely.

RULED OUT

- Clutter around the muscle. The muscle fills most of the frame, and the least cluttered pictures give the worst answers.

STILL OPEN — the actual root cause

- Is the fibre detector failing across the whole competition set, as it does on the picture in section 3? Needs checking on a sample, not one image.
- Or are the competition pictures stretched relative to their true proportions? That would shrink every measured angle. A stretch factor of about 1.35 lines the angles up, but we cannot yet separate a real stretch from the detector simply being wrong, so that number is not trusted.

NEXT STEP

- Score the fibre detector directly: on each competition picture, what fraction of its marks land inside the muscle? That separates the two explanations, needs no labels, and does not cost a submission.